Dominic Belyaev

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EDUCATION

Northwestern University - Evanston, IL
B.S. Mechanical Engineering (Aerospace), June 2028 | M.S. Aerospace Engineering (June 2029) - **GPA: 3.4**
Relevant Coursework: Mechanics of Materials, Thermodynamics, Stress Analysis, Fluid Mechanics, Materials Science, Electronics Design

WORK EXPERIENCE

Levanta Technologies **Diamondhead, MS**
Mechanical Engineering Intern **Jun - Aug 2026**
- Built a retractable keel fin system for a USAF/Navy-backed drone to stay rigid at 50 knots and fold 80° for ship-deck recovery
- Ran Ansys Fluent CFD to refine drag coefficient estimates from 0.10 to 0.011, cutting predicted fin hinge moment 9×
- Sized torsion springs with Shigley's method and iterated 3 printed prototypes in Fusion to a final redundant twin-fin design
- Fabricated plug & mold tooling & structural parts, helping to build a full-length carbon bottom skin that cut total weight 5%

Meemo (Northwestern Garage Startup Incubator) **Evanston, Illinois**
Industrial Design Intern **Jun - Jul 2025**
- Designed a 3-button actuation mechanism for a parent–child locating device, matching buttons to tactile PCB switches
- Packaged an RGB status indicator into the housing and verified plunger travel and component clearances in SOLIDWORKS

PROJECTS

Carbon Fiber Nosecone & Telemetry Rocket Build **May - Aug 2026**
- Designed & manufactured carbon fiber nosecone: one-shot layup in a printed split female mold, 47% lighter than fiberglass
- Designed & 3D printed avionics housing, sizing parachute load path for 300 N opening shock, surviving flight loads twice
- Hand-soldered and integrated a Pico 2 flight computer (IMU + barometer) into the avionics bay, logging 100 Hz flight data
- Used captured flight data to trace flight 1's failure root cause to parachute packing, then corrected it and successfully reflew
- Directed an AI-assisted 3D reconstruction of both flights from raw IMU data, matching the barometer at apogee within 3.3 m

Bicycle Brake Caliper Design **Mar - Jun 2026**
- Modeled & assembled caliper arms in NX, checking each against Nylon-12's 50 MPa ultimate strength at a 1.2 safety factor
- Cut critical-arm peak stress by 16% & raised deflection safety factor to 1.45 via an FEA & topology optimization redesign
- Tested brakes per ISO 4210-2 (official safety requirements for bikes), beating the 7.62 m stopping-distance spec at 5.33 m

IREC Rocket Payload Battery Sled (Northwestern U. Space Technology & Rocketry Society) **Jan 2026 - Present**
- Designed a compact battery sled in Onshape for integration into the IREC 25-26 competition rocket's avionics bay
- Integrated sled to secure electronics against loads in a ~10,000ft apogee flight using heat-set inserts & anti-vibration fasteners

DC Motor Control System **Jan - Mar 2026**
- Built a dual-loop PID control system (1 kHz current, 200 Hz position) using C on a Raspberry Pi for a brushed DC motor
- Achieved <25 mA average current tracking error and <9° average position tracking error through iterative PID tuning
- Developed a Python serial interface with 15+ real-time commands for trajectory generation, gain tuning, and data plotting

NAR Level 1 Rocket Certification **Apr - Jun 2026**
- Earned NAR Level 1 certification, building and flying an H182-powered rocket with single-deploy parachute recovery
- Modeled 2,460 ft apogee and a 1.6-cal stability margin in OpenRocket and tuned ejection delay to the simulated optimum
- Assembled a 3-inch airframe using epoxy, jig-aligning fins, & reinforcing the roots w/ carbon-reinforced epoxy fillets

MagClean **Mar - Jun 2025**
- Built a magnet-assisted whiteboard cleaner on a manual mill, band saw, and drill press; deployed in the Ford Design Center
- Cut full-board cleaning time ~50% vs standard erasers, tuning cleaning pressure and reach through user testing across heights

LEADERSHIP & ACTIVITIES

Applied Technologies Club (Co-Founder) **Dec 2025 - Present**
- Co-founded student organization focused on student-chosen & student-led hands-on robotics, defense, & engineering projects

Northwestern University Men's Rugby Club (Member) **Sep 2024 - Present**

Lambda Chi Alpha Fraternity (Social Chair) **Mar 2025 - May 2026**

SKILLS

Software: SOLIDWORKS, NX, Onshape, Fusion, Ansys Mechanical, Ansys Fluent, Python, C/C++, MATLAB, Microsoft 365 Suite
Design & Analysis: FEA/CFD, GD&T, Topology Optimization, Stress Analysis, Rapid Prototyping, Structures/Electronics Design
Fabrication: 3D Printing, Composites Manufacturing, Lathe, Mill, Laser Cutter, Sheet Metal Tools, Bandsaws, Soldering